

September 4, 2025

CURRICULUM VITAE – Abridged Version

Leroy L. Jia

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Citizenship: USA

Positions Held

Mathematician, Applied and Computational Mathematics Division, NIST 2023–
Flatiron Research Fellow, Center for Computational Biology, Flatiron Institute 2018–2022
Supervisor: Prof. Michael Shelley

Visiting Positions

Visitor, Courant Institute of Mathematical Sciences, New York University 2019–2020
Visiting Scientist, Department of Physics, Brown University 2018–2019
Postdoctoral Research Associate, School of Engineering, Brown University 2018
Supervisor: Prof. Thomas Powers (1-month appointment)

Education

Ph.D. Applied Mathematics, Brown University 2018
Advisors: Profs. Thomas Powers & Robert Pelcovits
Thesis: “Geometry and Mechanics of Self-Assembling Colloidal Membranes”
Sc.M. Engineering, Brown University 2017
Sc.M. Applied Mathematics, Brown University 2013
B.S. Physics, Georgia Institute of Technology 2012
B.S. Applied Mathematics, Georgia Institute of Technology 2012
Certificate in I/O Psychology, Georgia Institute of Technology 2012

Awards and Honors

Sigma Xi, Brown University Chapter 2018
Outstanding Teaching Assistant, Brown University 2014
School of Mathematics MCTP Scholarship, Georgia Institute of Technology 2012, 2011
Hitohiro Fukuyo Memorial Award, Georgia Institute of Technology 2011

Research Interests

Applied mathematics: analytical & computational modeling, PDE, asymptotics, geometry
Theory of soft matter: membranes, fluid interfaces, colloids, liquid crystals, active matter
Biomechanical engineering: complex fluids, elasticity, self-assembly, bio-inspired materials

Publications

9. X. Liu, N. A. Patankar, L. L. Jia. “A Plateau problem for membranes.” In preparation, draft available upon request (2025).
8. G. Chickering, L. L. Jia, M. DiSalvo, M. A. Catterton, P. N. Patrone, E. M. Darling, G. A. Cooksey. “Calibration and measurement of single-cell elastic modulus by time-of-flight and fluorescence signal analysis in a serial microfluidic cytometer.” Submitted, draft available upon request (2025)
7. R. Adkins, J. Robaszewski, S. Shin, F. Brauns, L. L. Jia, A. Khanra, P. Sharma, R. A. Pelcovits, T. R. Powers, Z. Dogic. “Topology and kinetic pathways of colloidosome assembly and disassembly.” *Proceedings of the National Academy of Sciences of the USA* **112**(36) (2025) e2427024122.
6. L. L. Jia, W. T. M. Irvine, M. J. Shelley. “Incompressible active phases at an interface. Part 1. Formulation and axisymmetric odd flows.” *Journal of Fluid Mechanics* **951** (2022) A36.
5. L. L. Jia, M. J. Shelley. “The role of monolayer viscosity in Langmuir film hole closure dynamics.” *Journal of Fluid Mechanics* **948** (2022) A1.
4. A. Khanra, L. L. Jia, N. P. Mitchell, A. Balchunas, R. A. Pelcovits, T. R. Powers, Z. Dogic, P. Sharma. “Controlling the shape and topology of two-component colloidal membranes.” *Proceedings of the National Academy of Sciences of the USA* **119**(32) (2022) e2204453119.
3. L. L. Jia, S. Pei, R. A. Pelcovits, T. R. Powers. “Axisymmetric membranes under external force: buckling, minimal surfaces, and tethers.” *Soft Matter* **17** (2021) 7268. Selected as Front Cover.
2. A. Balchunas, L. L. Jia, M. J. Zakhary, J. Robaszewski, T. Gibaud, Z. Dogic, R. A. Pelcovits, T. R. Powers. “Force-induced formation of twisted chiral ribbons.” *Physical Review Letters* **125** (2020) 018002.
1. L. L. Jia, M. J. Zakhary, Z. Dogic, R. A. Pelcovits, T. R. Powers. “Chiral edge fluctuations of colloidal membranes.” *Physical Review E* **95** (2017) 060701(R).

In the News

1. L. L. Jia and X. Liu. “Mathematical Models Can Help Us Understand – and Possibly Treat – Complex Diseases.” NIST Taking Measure Blog (2025).

Recent Presentations (from the last 4 years)

- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2025)
- Invited talk, SIAM/CAIMS Annual Meeting (July 2025)
- Poster, NIST ITL Science Day (December 2024)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2024)
- Poster, International Soft Matter Conference (July 2024)

- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2023)
- Seminar, National Institute of Standards and Technology (February 2023)
- Seminar, University of Alabama at Birmingham (February 2023)
- Seminar, University of Pennsylvania (January 2023)
- Seminar, National Security Agency (October 2022)
- Seminar, Flatiron Institute (October 2022)
- Seminar, Manhattan College (August 2022)
- Seminar, High Point University (July 2022)
- Seminar, Soft, Living, Active, and Adaptive Matter Seminar Series (June 2022)
- Poster, Mechanics of Life Workshop (May 2022)
- Seminar, The Cooper Union (April 2022)
- Seminar, Northwestern University (February 2022)
- Contributed talk, APS Division of Fluid Dynamics Meeting (November 2021)
- Seminar, Flatiron Institute (September 2021)
- Contributed talk, APS March Meeting (March 2021)

Teaching Experience

- Lecturer, NIST ACMD Reading Group (January 2024–June 2024). Topics included the calculus of variations, differential geometry, and theoretical mechanics.
- Instructor, North Carolina School of Science and Mathematics (Spring 2023). Taught biomedical engineering.
- Instructor, Division of Applied Mathematics, Brown University (Summer 2017). Taught introductory scientific computing.
- Teaching Consultant, Sheridan Center for Teaching and Learning, Brown University (Fall 2017–Spring 2018). Obtained Sheridan Center Reflective Teaching, Course Design, and Teaching Consultant Certificates.
- Graduate Teaching Assistant, Division of Applied Mathematics, Brown University (Fall 2013–Spring 2015). Taught methods of applied mathematics.
- Content Instructor and Peer Mentor, New Scientist Program Catalyst, Brown University (Summer 2014). Taught complex numbers and linear stability theory.
- Undergraduate Teaching Assistant, School of Physics, Georgia Institute of Technology (Spring 2009–Spring 2012). Taught introductory mechanics and electromagnetism.
- Undergraduate Teaching Assistant, School of Mathematics, Georgia Institute of Technology (Fall 2011–Spring 2012). Taught calculus and linear algebra.
- Undergraduate Grader, School of Mathematics, Georgia Institute of Technology (Fall 2010–Spring 2012). Graded for partial differential equations, abstract algebra, and real analysis courses.

Students Trained

- Xinyi Liu, Guest Researcher at NIST (Spring 2024–present). Co-advised with N. A. Patankar. Xinyi is a Ph.D. student in the Department of Engineering Sciences and Applied Mathematics at Northwestern University.
- Sydney Lee, SURF student at NIST (Summer 2024). Co-advised with M. DiSalvo. Sydney is an undergraduate student at Texas A&M University.
- Steven Pei, undergraduate physics student at Brown University (Fall 2016–Fall 2018). Co-advised with T. R. Powers.

Recent Service (from the last 4 years)

- Co-chair (w/ Z. Grey), NIST ACMD Seminar Series (Summer 2024–present)
- Events Committee, Association of NIST Asian-Pacific Americans (Fall 2023–present) & Secretary, Association of NIST Asian-Pacific Americans (Fall 2024–present)
- Georgia Tech Alumni Association Mentor (Fall 2024–present)
- Reviewer, NIST Small Business Innovation Research grant proposals (Spring 2024)
- Member, Simons Foundation AAPI Employee Resource Group (Winter 2021–Fall 2022)
- Professional Development Committee, Flatiron Institute (Winter 2021–Fall 2022)
- Referee, Journal of Fluid Mechanics ($\times 5$)

Personal

- Languages: English (native), Mandarin Chinese (fluent), Japanese (conversational)
- Memberships: APS (DFD & DSOF), SIAM

References

Mr. Joseph Browne (teaching reference)

Research Training Manager, University of Massachusetts at Boston

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(617)-229-5639

Prof. Zvonimir Dogic (frequent collaborator)

Dept. of Physics & Dept. of Biomolecular Science and Engineering, UC at Santa Barbara

Martin A. Fisher School of Physics, Brandeis University

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Prof. Robert Pelcovits (thesis advisor)

Department of Physics, Brown University

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Prof. Thomas Powers (thesis advisor)

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Prof. Michael Shelley (postdoctoral supervisor)

Center for Computational Biology, Flatiron Institute

Courant Institute of Mathematical Sciences, New York University

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